PLC PRACTICE PROBLEM PACK

30 Ladder Logic & Industrial Automation Exercises

Level 1 Fundamentals • 10 exercises	**Level 2** Intermediate machines • 10 exercises	**Level 3** Industrial / capstone • 10 exercises

Purpose: Solve the problems yourself before looking for a solution. Each exercise is written as a small industrial-control specification: identify the sequence, map the I/O, choose internal bits/timers/counters, write the ladder logic, and test edge cases.

Recommended workflow: 1) Read the machine description → 2) draw the sequence/state diagram → 3) create the I/O table → 4) define internal memory → 5) write ladder logic → 6) simulate → 7) test normal, abnormal and reset conditions.

Prepared as an original practice set inspired by common PLC training exercises.

How to use this pack

This is a **problem-only** workbook. It intentionally does not provide ladder solutions. For each problem, create your own solution in the PLC platform or simulator you are learning.

Step	What to produce
1	Machine sequence / state diagram
2	PLC I/O address list
3	Internal bits, timers, counters and registers
4	Ladder logic / FBD / SFC implementation
5	Simulation test cases
6	Fault, reset and edge-case tests
7	Short explanation of design decisions

Important: Real industrial safety functions such as emergency stops, guarding and safety-rated interlocks should not be implemented solely as ordinary PLC logic in a real machine. In these exercises, safety inputs are included for control-system design practice.

Level 1 — Fundamentals

Start/stop circuits, seal-in logic, timers, counters, sensors, simple sequencing and basic pneumatic control.

Topics: seal-in • TON • CTU • sensors • simple sequences

Level 2 — Intermediate

Machine sequences, batching, sorting, mode control, shift registers, counters, interlocks and fault handling.

Topics: state sequencing • shift registers • batch control • interlocks • manual/auto

Level 3 — Industrial / Capstone

Encoder tracking, analog scaling, VFD control, diagnostics, recipes, state machines and integrated production lines.

Topics: analog • encoder • VFD • diagnostics • recipes • capstone integration

1. Motor Start/Stop with Seal-In

Level 1 • Basic discrete control

Problem: A conveyor motor is controlled from a START pushbutton and STOP pushbutton. Pressing START should run the motor continuously after the button is released. Pressing STOP must stop it.

Machine / signal sketch

PB	START
PB	STOP
M	MOTOR

I/O specification

Tag	Type	Description
I0.0	Input	START pushbutton — NO
I0.1	Input	STOP pushbutton — NC
Q0.0	Output	Conveyor motor

Requirements

- Create a seal-in/holding circuit.
- The motor must remain ON after START is released.
- STOP must have priority over START.
- Do not use a timer or counter.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

2. Jog/Run Conveyor

Level 1 • Basic discrete control

Problem: A conveyor has RUN and JOG pushbuttons. RUN latches the conveyor ON. JOG runs the conveyor only while the JOG button is held.

Machine / signal sketch

PB	RUN
PB	JOG
PB	STOP
M	MOTOR

I/O specification

Tag	Type	Description
I0.0	Input	RUN pushbutton
I0.1	Input	STOP pushbutton
I0.2	Input	JOG pushbutton
Q0.0	Output	Conveyor motor

Requirements

- Implement RUN as a maintained mode.
- JOG must not latch.
- STOP must stop both RUN and JOG operation.
- Prevent RUN and JOG logic from creating an unsafe conflict.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

3. Two-Hand Start

Level 1 • Safety/interlock logic

Problem: A small press requires two pushbuttons to be pressed within the same machine cycle to start a motor. For this exercise, both buttons must be ON at the same time for the motor to energize.

Machine / signal sketch

PB	LEFT
PB	RIGHT
PB	STOP
M	PRESS

I/O specification

Tag	Type	Description
I0.0	Input	LEFT start PB
I0.1	Input	RIGHT start PB
I0.2	Input	STOP PB — NC
Q0.0	Output	Press motor

Requirements

- The motor may run only when both start buttons are active.
- STOP must immediately remove the run command.
- Do not latch the motor unless you intentionally define a separate cycle latch.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

4. Motor with Run Indicator

Level 1 • Outputs

Problem: A conveyor motor has a green RUN lamp. The lamp must indicate the commanded running state of the motor.

Machine / signal sketch

PB	START
PB	STOP
M	MOTOR
LAMP	RUN

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP — NC
Q0.0	Output	Motor
Q0.1	Output	Green RUN lamp

Requirements

- Build the motor seal-in circuit.
- Drive the RUN lamp from the appropriate PLC logic state.
- The lamp must turn OFF when the motor is stopped.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

5. Delayed Conveyor Start

Level 1 • TON timer

Problem: When START is pressed, a conveyor must wait 5 seconds before starting. A READY lamp should indicate that the start request has been accepted but the delay is still active.

Machine / signal sketch

PB	START
PB	STOP
TON 5 s	T1
M	MOTOR
LAMP	READY

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP — NC
Q0.0	Output	Conveyor motor
Q0.1	Output	READY lamp

Requirements

- Use a TON timer.
- The motor starts only after the full 5-second delay.
- STOP must cancel the pending start.
- READY should indicate the timing/request state.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

6. Automatic Motor Stop Timer

Level 1 • TON timer

Problem: Pressing START runs a conveyor for exactly 10 seconds, then the conveyor stops automatically. A STOP button must also stop it early.

Machine / signal sketch

PB	START
PB	STOP
TON 10 s	T1
M	MOTOR

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP — NC
Q0.0	Output	Conveyor motor

Requirements

• Create a 10-second timed run.
• Do not allow the timer to restart continuously while START remains pressed unless your design explicitly handles this.
• STOP must reset the cycle.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

7. Box Counter

Level 1 • CTU counter

Problem: Boxes pass a photoelectric sensor. Count each box and stop the conveyor when 10 boxes have been counted. RESET clears the count and allows a new batch.

Machine / signal sketch

PE	SENSOR
PB	RESET
PB	START
PB	STOP
M	MOTOR
CTU 10	C1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	Box sensor
I0.3	Input	RESET
Q0.0	Output	Conveyor motor

Requirements

- Count one event per box, not every PLC scan while the sensor is ON.
- Stop the conveyor at preset 10.
- RESET clears the counter.
- Define what happens if START is pressed after the batch is complete.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

8. Tank Fill with High-Level Stop

Level 1 • Sensors

Problem: A tank is filled by a pump. Press START to enable filling. The pump must stop when the HIGH level switch becomes active. LOW level indicates the tank is below its lower operating point.

Machine / signal sketch

LEVEL	LOW
LEVEL	HIGH
PB	START
PB	STOP
M	PUMP

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	LOW level
I0.3	Input	HIGH level
Q0.0	Output	Fill pump

Requirements

- Create a maintained fill command.
- The pump must not run when HIGH is active.
- Use LOW as a permissive/indication according to your sequence.
- Document your assumptions about what happens between LOW and HIGH.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

9. Traffic Light Cycle

Level 1 • Timers/sequencing

Problem: A simple traffic light operates continuously after START: GREEN for 10 s, then AMBER for 3 s, then RED for 10 s, then repeat. STOP returns the system to an idle state.

Machine / signal sketch

PB	START
PB	STOP
LAMP	GREEN
LAMP	AMBER
LAMP	RED

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
Q0.0	Output	GREEN
Q0.1	Output	AMBER
Q0.2	Output	RED

Requirements

- Implement the sequence using timers and internal states/bits as needed.
- Only one light may be ON at a time.
- STOP must terminate the sequence cleanly.
- Choose and document how the cycle initializes.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

10. Pneumatic Cylinder Extend/Retract

Level 1 • Sequential basics

Problem: A pneumatic cylinder has EXTEND and RETRACT solenoids plus two limit switches. Press START to extend the cylinder; when fully extended it should retract automatically; when fully retracted the cycle is complete.

Machine / signal sketch

PB	START
LS	EXT
LS	RET
SOL	EXTEND
SOL	RETRACT

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	EXTENDED limit
I0.2	Input	RETRACTED limit
Q0.0	Output	Extend solenoid
Q0.1	Output	Retract solenoid

Requirements

- Never command extend and retract simultaneously.
- Use the limit switches to transition between stages.
- Prevent the sequence from restarting unexpectedly.
- Add a reset/fault assumption if you believe it is necessary.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

11. Bottle Filling Station

Level 2 • Timers + sensors

Problem: Empty bottles arrive on a conveyor. Sensor B detects a bottle at the filling position. Stop the conveyor, open the fill valve for 4 seconds, close the valve, then restart the conveyor.

Machine / signal sketch

PE	BOTTLE
PB	START
PB	STOP
M	CONVEYOR
SOL	VALVE
TON 4 s	T1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	Bottle-at-fill sensor
Q0.0	Output	Conveyor
Q0.1	Output	Fill valve

Requirements

- Stop the conveyor while filling.
- Open the valve only when a bottle is correctly positioned.
- Restart after the 4-second fill.
- Prevent multiple fill cycles from being triggered by one bottle.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

12. Bottle Capper

Level 2 • Sequential control

Problem: A bottle enters a capping station. The conveyor stops when the bottle is detected. A capper cylinder extends, waits for the EXTENDED switch, holds for 1 second, then retracts. Once RETRACTED is detected, the conveyor restarts.

Machine / signal sketch

PE	BOTTLE
LS	EXT
LS	RET
M	CONVEYOR
SOL	EXTEND
SOL	RETRACT

I/O specification

Tag	Type	Description
I0.0	Input	Bottle sensor
I0.1	Input	EXTENDED limit
I0.2	Input	RETRACTED limit
I0.3	Input	START
Q0.0	Output	Conveyor
Q0.1	Output	Extend
Q0.2	Output	Retract

Requirements

- Build distinct sequence states.
- The conveyor must remain stopped during capping.
- Use the limit switches for confirmation.
- Include a 1-second hold at full extension.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

13. Conveyor Sorting by Sensor

Level 2 • Routing

Problem: Products travel on a conveyor. When a product is detected by the classification sensor, a diverter cylinder must activate for 1 second to send the product to Lane 2. Otherwise the product continues to Lane 1.

Machine / signal sketch

PE	PRODUCT
SENSOR	CLASS
M	CONVEYOR
SOL	DIVERTER
TON 1 s	T1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	Product sensor
I0.2	Input	Reject/classify sensor
Q0.0	Output	Conveyor
Q0.1	Output	Diverter

Requirements

- Only activate the diverter for classified products.
- Time the diverter for 1 second.
- Consider what happens if two products arrive close together.
- State your assumptions about minimum product spacing.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

14. Automatic Parking Barrier

Level 2 • Counters

Problem: A parking entrance has an entry sensor and exit sensor. The barrier opens when a vehicle is detected at the entrance and closes after the vehicle passes the barrier. Maintain an occupancy count with a maximum of 20 vehicles.

Machine / signal sketch

PE	ENTRY
PE	EXIT
LS	OPEN
LS	CLOSED
M	OPEN
M	CLOSE
CTU	C1

I/O specification

Tag	Type	Description
I0.0	Input	Entry sensor
I0.1	Input	Exit sensor
I0.2	Input	Barrier-open limit
I0.3	Input	Barrier-closed limit
Q0.0	Output	Barrier open motor
Q0.1	Output	Barrier close motor

Requirements

- Increment occupancy for each valid entry.
- Decrement occupancy for each valid exit.
- Do not allow entry when occupancy is 20.
- Use the limit switches to stop the barrier motor.
- Prevent double-counting when a sensor remains active.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

15. Tank Mixing Sequence

Level 2 • Multi-step sequence

Problem: A tank has a fill valve, mixer and drain valve. Press START: fill until HIGH level, mix for 15 seconds, then drain until LOW level. Repeat for three batches and stop.

Machine / signal sketch

LEVEL	LOW
LEVEL	HIGH
SOL	FILL
M	MIXER
SOL	DRAIN
TON 15 s	T1
CTU 3	C1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	LOW level
I0.2	Input	HIGH level
Q0.0	Output	Fill valve
Q0.1	Output	Mixer
Q0.2	Output	Drain valve

Requirements

- Implement FILL → MIX → DRAIN → next batch.
- Mixer runs only during MIX.
- Stop after three completed batches.
- Ensure fill and drain cannot be commanded together.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

16. Conveyor Reject with Shift Register

Level 2 • Shift register

Problem: Products enter a conveyor at Sensor B. Sensor A identifies a product that must be rejected. An encoder produces one pulse for each fixed conveyor movement. Track the reject decision in a shift register and activate an air blow-off when the marked product reaches the reject station.

Machine / signal sketch

PE	SENSOR A
PE	SENSOR B
PULSE	ENCODER
M	CONVEYOR
SOL	AIR
REG	SHIFT

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	Sensor A — reject detect
I0.3	Input	Sensor B — product detect
I0.4	Input	Encoder pulse
Q0.0	Output	Conveyor
Q0.1	Output	Reject air

Requirements

- Create a shift-register representation of conveyor positions.
- Load a reject bit when the appropriate product enters.
- Shift only on encoder events.
- Activate the reject output only when the reject bit reaches the reject position.
- Define the number of positions between detection and rejection.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

17. Automatic Gate with Vehicle Safety

Level 2 • Interlocks

Problem: A gate opens when a vehicle approaches. It remains open while the vehicle is under the gate and closes after the vehicle has passed. A safety beam must prevent closing while blocked.

Machine / signal sketch

PE	APPROACH
PE	UNDER
SAFETY	BEAM
LS	OPEN
LS	CLOSED
M	OPEN
M	CLOSE

I/O specification

Tag	Type	Description
I0.0	Input	Approach sensor
I0.1	Input	Under-gate sensor
I0.2	Input	Safety beam
I0.3	Input	OPEN limit
I0.4	Input	CLOSED limit
Q0.0	Output	Open motor
Q0.1	Output	Close motor

Requirements

- Never command both motor directions together.
- The gate must not close while the safety beam is blocked.
- Use the limit switches to stop movement.
- Define a reasonable close delay or closing condition.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

18. Dual Pump Alternation

Level 2 • Alternating duty

Problem: A tank uses two pumps. When the level is HIGH, a pump must run until the level returns to LOW. Alternate the duty pump after each completed pumping cycle. If the duty pump is unavailable, the standby pump should run.

Machine / signal sketch

LEVEL	HIGH
LEVEL	LOW
INPUT	P1 FAULT
INPUT	P2 FAULT
M	P1
M	P2

I/O specification

Tag	Type	Description
I0.0	Input	HIGH level
I0.1	Input	LOW level
I0.2	Input	Pump 1 fault
I0.3	Input	Pump 2 fault
Q0.0	Output	Pump 1
Q0.1	Output	Pump 2

Requirements

- Alternate duty between cycles.
- A failed duty pump should transfer to the other pump if available.
- Do not run both pumps unless your sequence explicitly requires it.
- Define startup duty selection.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

19. Bottle Counting and Batch Release

Level 2 • Counter + sequence

Problem: Count 12 bottles on a conveyor. When 12 are counted, stop the conveyor and run a batch-release cylinder for 2 seconds. Reset the count and start the next batch.

Machine / signal sketch

PE	BOTTLE
M	CONVEYOR
SOL	RELEASE
CTU 12	C1
TON 2 s	T1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	Bottle sensor
Q0.0	Output	Conveyor
Q0.1	Output	Release cylinder

Requirements

- Count one per bottle.
- Stop at 12.
- Activate release for 2 seconds.
- Reset the batch count only at the correct point in the sequence.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

20. Manual/Automatic Conveyor Station

Level 2 • Mode selection

Problem: A conveyor station has MANUAL and AUTO modes. In MANUAL, an operator can jog the conveyor. In AUTO, products are detected and the conveyor runs a programmed sequence. A mode switch selects the operating mode.

Machine / signal sketch

SELECT	AUTO
PB	START
PB	STOP
PB	JOG
PE	PRODUCT
M	MOTOR

I/O specification

Tag	Type	Description
I0.0	Input	AUTO mode selector
I0.1	Input	START
I0.2	Input	STOP
I0.3	Input	JOG
I0.4	Input	Product sensor
Q0.0	Output	Conveyor

Requirements

- Separate manual and automatic control paths.
- JOG must work only in MANUAL.
- Automatic operation must not be affected by JOG.
- Define what happens when the operator changes mode while the conveyor is running.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

21. Encoder-Based Conveyor Positioning

Level 3 • Encoder + shift register

Problem: A conveyor uses an encoder to represent distance. Products are detected at the load point and must be tracked through 16 virtual positions. Product type is captured at entry and used to trigger different outputs at the discharge station.

Machine / signal sketch

PE	ENTRY
SENSOR	TYPE A
PULSE	ENCODER
SOL	A DIVERT
SOL	B DIVERT
REG	SHIFT

I/O specification

Tag	Type	Description
I0.0	Input	Product entry sensor
I0.1	Input	Type A sensor
I0.2	Input	Encoder pulse
I0.3	Input	Reset
Q0.0	Output	Type A diverter
Q0.1	Output	Type B diverter

Requirements

- Use encoder pulses as the shift event.
- Store product classification with the product position.
- Trigger the appropriate discharge output at position 16.
- Reset must clear all tracked positions.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

22. Two-Station Packaging Line

Level 3 • State machine

Problem: A product enters Station 1, where it is clamped and processed for 5 seconds. It then moves to Station 2, where it is processed for 8 seconds. The conveyor may move only when both stations are clear.

Machine / signal sketch

PE	S1 PRODUCT
PE	S2 PRODUCT
M	CONVEYOR
SOL	CLAMP
M	PROCESS
TON 5 s	T1
TON 8 s	T2

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	Stop
I0.2	Input	Station 1 product
I0.3	Input	Station 2 product
Q0.0	Output	Conveyor
Q0.1	Output	Station 1 clamp
Q0.2	Output	Station 2 process

Requirements

- Create explicit states for transport and processing.
- Prevent conveyor motion while a station is processing.
- Station 1 must complete before the product proceeds.
- Station 2 must complete before discharge.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

23. Analog Tank Level Control

Level 3 • Analog scaling

Problem: A tank has a 4–20 mA level transmitter representing 0–100%. The PLC should scale the raw analog value to engineering units. A pump is controlled to maintain the level between 40% and 60%.

Machine / signal sketch

4–20mA	LEVEL
SELECT	AUTO
M	PUMP

I/O specification

Tag	Type	Description
AI0	Analog Input	4–20 mA level transmitter
I0.0	Input	AUTO selector
Q0.0	Output	Pump

Requirements

- Define the raw input range and engineering-unit range.
- Implement scaling conceptually or with the PLC's available scaling instruction.
- Pump ON below 40%. Pump OFF above 60%.
- Explain how hysteresis prevents rapid pump cycling.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

24. Analog Pressure Alarm

Level 3 • Analog + alarms

Problem: A pressure transmitter provides 4–20 mA corresponding to 0–10 bar. Generate a HIGH alarm at 8 bar and a HIGH-HIGH trip at 9 bar. A reset is required for the latched trip.

Machine / signal sketch

4–20mA	PRESSURE
PB	RESET
ALARM	HIGH
LATCH	TRIP

I/O specification

Tag	Type	Description
AI0	Analog Input	4–20 mA pressure transmitter
I0.0	Input	RESET
Q0.0	Output	HIGH alarm
Q0.1	Output	TRIP

Requirements

- Scale the raw signal to bar.
- HIGH alarm activates at 8 bar.
- TRIP activates at 9 bar and latches.
- RESET should clear the trip only when pressure is below the defined safe threshold.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

25. VFD Conveyor Speed Selection

Level 3 • VFD control

Problem: A conveyor uses a VFD with three preset speeds: LOW, MEDIUM and HIGH. The operator selects a speed and starts the conveyor. A fault input from the VFD must stop the run command and illuminate a fault lamp.

Machine / signal sketch

PB	START
PB	STOP
SELECT	SPEED
INPUT	VFD FAULT
VFD	RUN
VFD	S1/S2
LAMP	FAULT

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	LOW speed
I0.3	Input	MED speed
I0.4	Input	HIGH speed
I0.5	Input	VFD fault
Q0.0	Output	VFD run
Q0.1	Output	Speed bit 1
Q0.2	Output	Speed bit 2
Q0.3	Output	Fault lamp

Requirements

- Implement mutually exclusive speed selection.
- VFD fault must remove RUN.
- Define the speed-bit truth table.
- Fault indication remains until the fault condition is cleared or reset according to your chosen design.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

26. Motor Interlock and Fault Reset

Level 3 • Industrial motor control

Problem: Two motors drive the same mechanism. Motor 2 may run only if Motor 1 is already running. Either motor's overload fault must stop both motors. A RESET clears a latched fault after the fault inputs are no longer active.

Machine / signal sketch

PB	START
PB	STOP
FAULT	M1 OL
FAULT	M2 OL
PB	RESET
M	M1
M	M2
LAMP	FAULT

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	M1 overload
I0.3	Input	M2 overload
I0.4	Input	RESET
Q0.0	Output	Motor 1
Q0.1	Output	Motor 2
Q0.2	Output	Fault lamp

Requirements

- Motor 2 requires Motor 1 running.
- Any overload stops both.
- Latch the fault condition.
- RESET is permitted only after active fault inputs are cleared.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

27. Automated Palletizer Sequence

Level 3 • Complex sequence

Problem: Boxes arrive on a conveyor. When four boxes have accumulated, stop the conveyor, extend a pusher, lower a lift, retract the pusher, raise the lift, then restart the conveyor for the next pallet.

Machine / signal sketch

PE	BOX
LS	PUSH EXT
LS	LIFT DOWN
LS	PUSH RET
LS	LIFT UP
M	CONVEYOR
SOL	PUSH
SOL	LIFT
CTU 4	C1

I/O specification

Tag	Type	Description
I0.0	Input	Box sensor
I0.1	Input	Pusher extended
I0.2	Input	Lift down
I0.3	Input	Pusher retracted
I0.4	Input	Lift up
Q0.0	Output	Conveyor
Q0.1	Output	Pusher extend
Q0.2	Output	Pusher retract
Q0.3	Output	Lift down
Q0.4	Output	Lift up

Requirements

- Count four boxes.
- Use state-based sequencing for the pallet operation.
- Never command opposing cylinder directions together.
- Restart box counting only after the pallet cycle is complete.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

28. Conveyor Reject with Fault Monitoring

Level 3 • Shift register + diagnostics

Problem: Extend the conveyor reject problem: track reject products using an encoder-based shift register. Monitor the reject air pressure switch and a reject-cylinder feedback sensor. If a reject command occurs but feedback is not received within 1 second, latch a fault and stop the conveyor.

Machine / signal sketch

PE	PRODUCT
SENSOR	REJECT
PULSE	ENCODER
PRESSURE	AIR OK
LS	FEEDBACK
SOL	REJECT
LAMP	FAULT
REG	SHIFT

I/O specification

Tag	Type	Description
I0.0	Input	Product sensor
I0.1	Input	Reject sensor
I0.2	Input	Encoder
I0.3	Input	Air pressure OK
I0.4	Input	Reject feedback
I0.5	Input	RESET
Q0.0	Output	Conveyor
Q0.1	Output	Reject valve
Q0.2	Output	Fault lamp

Requirements

- Track the reject decision to the reject position.
- Do not reject if the required air permissive is absent.
- After a reject command, require feedback within 1 second.
- Latch the fault and stop the conveyor if feedback fails.
- RESET clears the fault only after the fault condition is safe.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

29. Automatic Batch Mixing with Recipe Selection

Level 3 • Recipe/state control

Problem: The operator selects Recipe A or Recipe B. Recipe A requires 20 seconds mixing; Recipe B requires 35 seconds. Both recipes fill to HIGH, mix for their recipe time, drain to LOW, and repeat for two batches.

Machine / signal sketch

SELECT	RECIPE
LEVEL	LOW
LEVEL	HIGH
SOL	FILL
M	MIXER
SOL	DRAIN
TON	T1
CTU 2	C1

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	Recipe A
I0.3	Input	Recipe B
I0.4	Input	LOW level
I0.5	Input	HIGH level
Q0.0	Output	Fill
Q0.1	Output	Mixer
Q0.2	Output	Drain

Requirements

- Allow only one recipe to be active.
- Select the mixing time from the recipe.
- Run two complete batches.
- Do not allow the recipe selection to change mid-batch unless you explicitly design for it.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

30. Integrated Mini Production Line

Level 3 • Capstone

Problem: Design the PLC control for a mini production line: products enter on Conveyor 1, are classified, transferred to Conveyor 2, optionally rejected, counted into batches of 10, and stopped when the batch is complete. Include AUTO/MANUAL mode, emergency-stop permissive, motor interlocks, a reject shift register, encoder tracking, fault indication and RESET.

Machine / signal sketch

PB	START
PB	STOP
SELECT	MODE
OK	E-STOP
PE	PRODUCT
SENSOR	REJECT
PULSE	ENCODER
PB	RESET
M	CONV 1
M	CONV 2
SOL	REJECT
LAMP	FAULT
REG	SHIFT
CTU 10	COUNT

I/O specification

Tag	Type	Description
I0.0	Input	START
I0.1	Input	STOP
I0.2	Input	AUTO/MANUAL
I0.3	Input	E-STOP healthy
I0.4	Input	Product sensor
I0.5	Input	Reject sensor
I0.6	Input	Encoder
I0.7	Input	RESET
Q0.0	Output	Conveyor 1
Q0.1	Output	Conveyor 2
Q0.2	Output	Reject valve
Q0.3	Output	Batch complete lamp
Q0.4	Output	Fault lamp

Requirements

- Design the complete I/O and internal-bit architecture.
- Implement safe start permissives and motor interlocks.
- Separate MANUAL and AUTO behavior.
- Track rejects using encoder-based position logic.
- Count 10 completed products.
- Define fault conditions and reset behavior.
- Document the state/sequence diagram before writing ladder logic.
- Treat this as a full PLC design exercise: assumptions, I/O map, sequence, ladder, testing and fault cases.

Your work: Draw the sequence/state diagram, write the ladder logic, and list at least three test cases (normal operation, reset/stop, and a fault or edge case).

Final PLC Design Checklist

1. ■ Have I defined every physical input and output?
2. ■ Have I clearly defined NO/NC behavior for pushbuttons and sensors?
3. ■ Have I identified which conditions are permissives and which are commands?
4. ■ Can two conflicting outputs ever be energized at the same time?
5. ■ Do timers start and reset at the correct points?
6. ■ Do counters count one physical event rather than repeated PLC scans?
7. ■ What happens after STOP?
8. ■ What happens after RESET?
9. ■ What happens after a power cycle?
10. ■ What happens if a sensor is stuck ON?
11. ■ What happens if a required feedback signal never arrives?
12. ■ Have I tested the first scan / initial state?
13. ■ Have I tested the sequence at its transition boundaries?
14. ■ For analog signals, have I considered scaling, limits and hysteresis?
15. ■ For encoder/shift-register problems, have I defined exactly when a shift occurs?
16. ■ Have I documented assumptions that were not specified in the problem?

Suggested next step: Start with Exercises 1–10. Do not move to Level 2 until you can write the logic, explain every rung, and test the sequence without relying on a tutorial.